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Assignment h1
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Assignment h1 CyberScience

A)

I picked an article named “A Review on Penetration Testing for Privacy of Deep Learning Models” published in Computers, Materials & Continua in 2026 (JUFO level 1). https://doi.org/10.32604/cmc.2026.076358

I found the article from Google Scholar with the search term “intitle:review "penetration testing"". After going through some and finding a good looking article, I tested the publisher on JUFO-portal for it to have a JUFO rating. Repeated this a couple times and ended up in this article.

B)

Key areas of research

- Insufficiency of traditional penetration testing frameworks regarding data protection against deep learning model security.
- Different ways to attack against deep learning models
- Analyzing different methods of protection strengthening information security

What's widely agreed upon?

- Standardized testing methods are insufficient in testing unique vulnerabilities of AI models
- There is no single method yet to protect against threats to AI models
- Development of new protection methods is promising, but slow and challenging because of for example rapid development of attacks

What are the open questions?

- How to develop fully automated and regulated penetration testing frameworks specially for AI models?
- How to keep those frameworks updated to keep up with upcoming threats?

C)

Threat modeling quickrun

Totally made up case, small local bank, let’s say Southia Bank Oy.

- Ground jewels: **Money** - Own and customers moneys
  Data - Customer information (identity-, account information…)
  Reputation - There is no bank without trust
  Business continuity - Online banking and card payments cannot be stopped.

- Bad guys:
    - Who - International cyber criminal group specialized in financial crime. Chasing money.
    - How - Ransomware, email or deep fake scam.
    - Why - Small bank, smaller Cyber Security budget

- Risk 1:
    - Deep fake or email scam. Criminals try to convince bank employees to send money with the wrong identity.
    - Likelihood: 20% (Within a year)
    - Impact: 500 000 € (direct financial loss)
    - Value: 20% x 500 000 € = 100 000 €
- Risk 2:
    - Ransomware on systems, Online banking stopped
    - Likelihood: 5% (Within a year)
    - Impact: 10 000 000 €
    - Value: 5% x 10 000 000 € = 500 000 €

- Biggest risk: Ransomware paralyzing continuity / 500 000 €

- What shall we do (META):
    - **M**itigate, we cannot fully **e**liminate the risk, cyber criminals don't stop. We cannot **a**ccept because consequences are too high, and we cannot **t**ransfer the risk to anyone else, reputation and fines are ours.

    - We are mitigating the risk by isolating main banking functions to a separate environment. Office networks are separated from having direct functionality to that environment and personnel access rights are limited to a minimum. To ensure continuity of services, we introduce unchangeable backups.